

ASSIGNMENT # 02

Subject: **Artificial Intelligence**
 Program: MCA
 Due Date: **3-12-2024**

Sub Code: **22MC314**
 Semester: **03 (Batch-2023)**
 Full Marks: **50**

Answer all 10 Questions. Each question carries 5 marks.

Q No.	CO*	Level	(1) Remembering (2) Understanding (3) Applying (4) Analyzing (5) Evaluating (6) Creating
1	3	2	What is planning in AI? Describe its key components.
2	3	3	How is partial order planning different from total order planning? Explain partial order planning with the Blocks world problem.
3	3	5	State the reasons for uncertainty in an environment. Briefly explain different types of uncertainties that may affect problem solving in AI.
4	3	3	Explain how probabilistic reasoning can deal with uncertainties with an example.
5	3	4	What is a Markov process? Explain how Hidden Markov model can be used to handle uncertainties indicating each of the probabilities.
6	4	2	What is an Expert system? How is it different from a normal software? Describe two of the application areas of expert system.
7	4	4	Provide a block diagram of an expert system with all its components. Explain the role of each of the components.
8	4	5	Distinguish between forward chaining and backward chaining as the reasoning mechanisms for building an inference engine.
9	4	3	Critically analyze the knowledge engineering process for the development of an effective expert system.
10	4	4	What is meant by an expert system development life cycle? Explain each phase of the life cycle.